UNIT-1

Introduction and Classical Optimization Techniques

Optimization is the act of obtaining the best result under given circumstances. In design, construction, and maintenance of any engineering system, engineers have to take many technological and managerial decisions at several stages. The ultimate goal of all such decisions is either to minimize the effort required or to maximize the desired benefit. Since the effort required or the benefit desired in any practical situation can be expressed as a function of certain decision variables, *optimization* can be defined as the process of finding the conditions that give the maximum or minimum value of a function. It can be seen from Fig. 1.1 that if a point x∗ corresponds to the minimum value of function f (x), the same point also corresponds to the maximum value of the negative of the function, −f (x). Thus without loss of generality, optimization can be taken to mean minimization since the maximum of a function can be found by seeking the minimum of the negative of the same function.

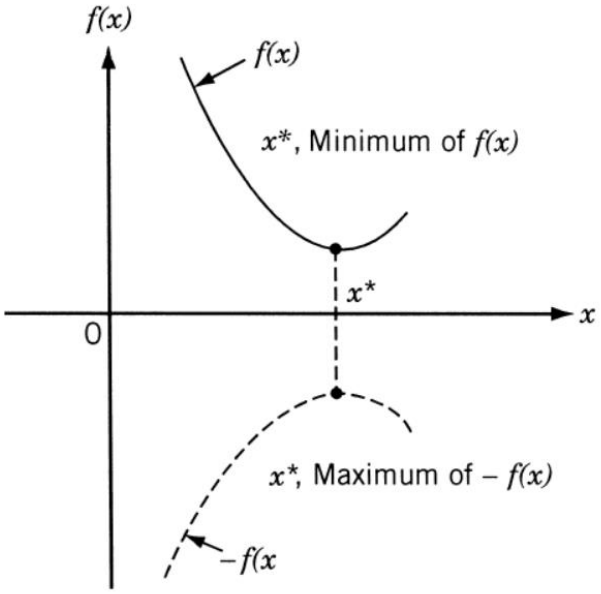


Figure 1.1 Minimum of $f(x)$ is same as maximum of $-f(x)$.

In addition, the following operations on the objective function will not change the

optimum solution x∗ (see Fig. 1.2):

1. Multiplication (or division) of f (x) by a positive constant c.

2. Addition (or subtraction) of a positive constant c to (or from) f (x).

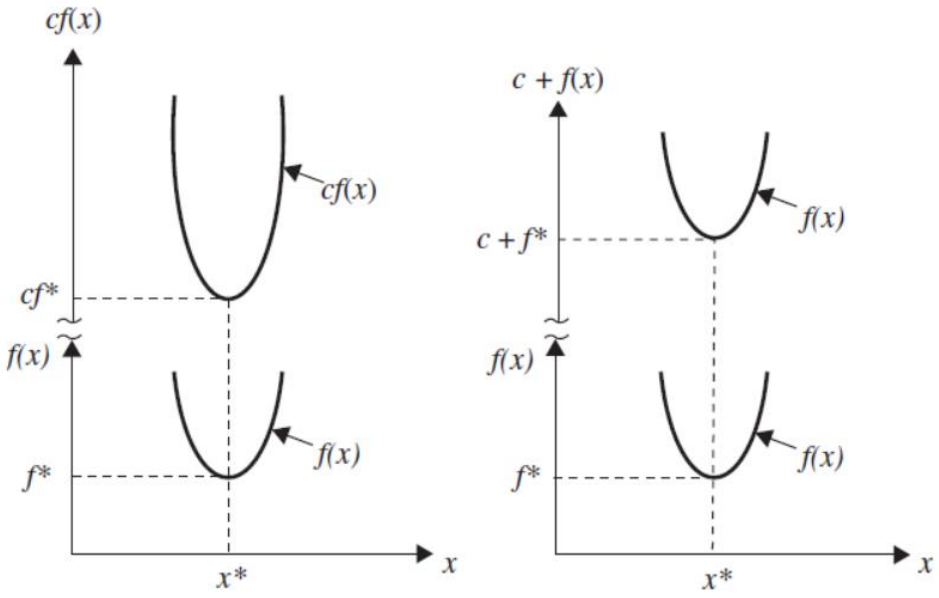


Figure 1.2 Optimum solution of $cf(x)$ or $c + f(x)$ same as that of $f(x)$.

There is no single method available for solving all optimization problems efficiently. Hence a number of optimization methods have been developed for solving different types of optimization problems. The optimum seeking methods are also known as *mathematical programming techniques* and are generally studied as a part of operations research. *Operations research* is a branch of mathematics concerned with the application of scientific methods and techniques to decision making problems and with establishing the best or optimal solutions.

STATEMENT OF AN OPTIMIZATION PROBLEM

An optimization or a mathematical programming problem can be stated as follows.

$$\text{Find } \mathbf{X} = \begin{Bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{Bmatrix} \text{ which minimizes } f(\mathbf{X})$$

subject to the constraints

$$g_j(\mathbf{X}) \leq 0, \qquad j = 1, 2, \ldots, m$$
$$l_j(\mathbf{X}) = 0, \qquad j = 1, 2, \ldots, p$$

where **X** is an n-dimensional vector called the *design vector*, f (**X**) is termed the *objective function*, and gj (**X**) and lj (**X**) are known as *inequality* and *equality* constraints, respectively. The number of variables n and the number of constraints m and/or p need not be related in any way. The problem stated in above Equation is called a *constrained optimization problem*.† Some optimization problems do not involve any constraints and can be stated as

$$\text{Find } \mathbf{X} = \begin{Bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{Bmatrix} \text{ which minimizes } f(\mathbf{X})$$

Such problems are called *unconstrained optimization problems*.

Design Vector

Any engineering system or component is defined by a set of quantities some of which are viewed as variables during the design process. In general, certain quantities are usually fixed at the outset and these are called *preassigned parameters*. All the other quantities are treated as variables in the design process and are called *design* or *decision variables* xi , i = 1, 2, . . . , n. The design variables are collectively represented as a design vector **X** = {x1, x2, . . . , xn}T.

Design Constraints

In many practical problems, the design variables cannot be chosen arbitrarily; rather,they have to satisfy certain specified functional and other requirements. The restrictions that must be satisfied to produce an acceptable design are collectively called *design constraints*. Constraints that represent limitations on the behaviour or performance of the system are termed *behavior* or *functional constraints*. Constraints that represent physical limitations on design variables, such as availability, fabricability, and transportability, are known as *geometric* or *side constraints*.

Constraint Surface

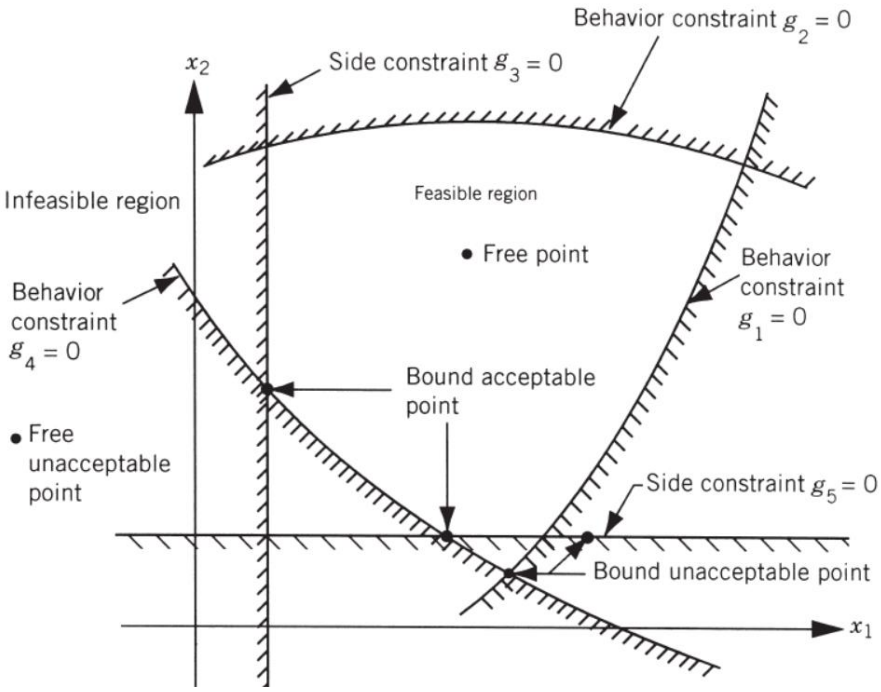
Figure shows a hypothetical two-dimensional design space where the infeasible region is indicated by hatched lines. A design point that lies on one or more than one constraint surface is called a *bound point* , and the associated constraint is called an *active constraint* . Design points that do not lie on any constraint surface are known as *free points*. Depending on whether a particular design point belongs to the acceptable or unacceptable region, it can be identified as one of the following four types:

1. Free and acceptable point

2. Free and unacceptable point

3. Bound and acceptable point

4. Bound and unacceptable point

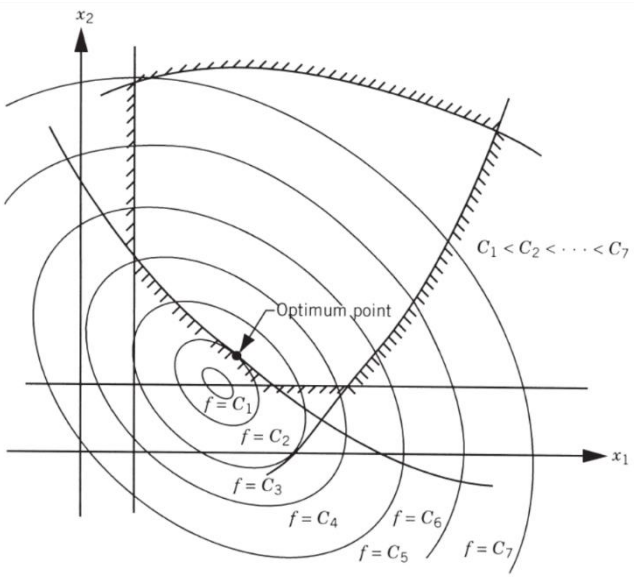


Objective Function

The conventional design procedures aim at finding an acceptable or adequate design that merely satisfies the functional and other requirements of the problem. In general, there will be more than one acceptable design, and the purpose of optimization is to choose the best one of the many acceptable designs available. Thus a criterion has to be chosen for comparing the different alternative acceptable designs and for selecting the best one. The criterion with respect to which the design is optimized, when expressed as a function of the design variables, is known as the *criterion* or *merit* or *objective function*.

Objective Function Surfaces

The locus of all points satisfying f (**X**) = C = constant forms a hypersurface in the design space, and each value of C corresponds to a different member of a family of surfaces. These surfaces, called *objective function surfaces*, are shown in a hypothetical two-dimensional design space in Figure



Classical Optimizati

Single Variable Optimization

Consider the function $f(x)$ defined for$axb\leq\leq$. To find the value of $x^* \in$ such that $x = x[,]ab^*$ maximizes $f(x)$ we need to solve a *single-variable optimization* problem. We have the following theorems to understand the necessary and sufficient conditions for the relative maximum of a function of a single variable.

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Necessary condition: For a single variable function $f(x)$ defined for x which has a relative maximum at $x = x[,]ab\in^*$, $x^* \in$ if the derivative exists as a finite number at $x = x^*$ then $f'(x^*) = 0$. This can be understood from the following.

Sufficient condition: For the same function stated above let $f'(x^*) = f''(x^*) = \ldots = f^{(n-1)}(x^*) = 0$, but $f^{(n)}(x^*)$ 0, then it can be said that $f(x^*)$ is (a) a minimum value of $f(x)$ if $f^{(n)}(x^*) > 0$ and n is even; (b) a maximum value of $f(x)$ if $f^{(n)}(x^*) < 0$ and n is even; (c) neither a maximum or a minimum if n is odd.

Example 1.

Find the optimum value of the function $f(x) = x^2 + 3x - 5$ and also state if the function attains a maximum or a minimum.

Solution

$f'(x) = 2x + 3 = 0$ for maxima or minima.

or $x^* = -3/2$

$f''(x^*) = 2$ which is positive hence the point $x^* = -3/2$ is a point of minima and the function attains a minimum value of -29/4 at this point.

Example 2.

Find the optimum value of the function $f(x) = (x-2)^4$ and also state if the function attains a maximum or a minimum.

Solution

$f'(x) = 4(x-2)^3 = 0$ for maxima or minima.

or $x = x^* = 2$ for maxima or minima.

$f''(x^*) = 12(x^*-2)^2 = 0$ at $x^* = 2$

$f'''(x^*) = 24(x^*-2) = 0$ at $x^* = 2$

$f'''(x^*) = 24$ at $x^* = 2$

Hence $f^n(x)$ is positive and n is even hence the point $x = x^* = 2$ is a point of minimum and the function attains a minimum value of 0 at this point.

Example 3.

Analyze the function $f(x) = 12x^5 - 45x^4 + 40x^3 + 5$ and classify the stationary points as maxima, minima and points of inflection.

Solution

$$f'(x) = 60x^4 - 180x^3 + 120x^2 = 0$$
$$\Rightarrow x^4 - 3x^3 + 2x^2 = 0$$
or $$x = 0, 1, 2$$

Consider the point $x = x^* = 0$

$$f'(x^*) = 240(x^*)^3 - 540(x^*)^2 + 240x^* = 0 \text{ at } x^* = 0$$

$$f''(x^*) = 720(x^*)^2 - 1080x^* + 240 = 240 \text{ at } x^* = 0$$

Since the third derivative is non-zero, $x = x^* = 0$ is neither a point of maximum or minimum but it is a point of inflection.

Consider $x = x^* = 1$

$$f'(x^*) = 240(x^*)^3 - 540(x^*)^2 + 240x^* = -60 \text{ at } x^* = 1$$

Since the second derivative is negative the point $x = x^* = 1$ is a point of local maxima with a maximum value of $f(x) = 12 - 45 + 40 + 5 = 12$

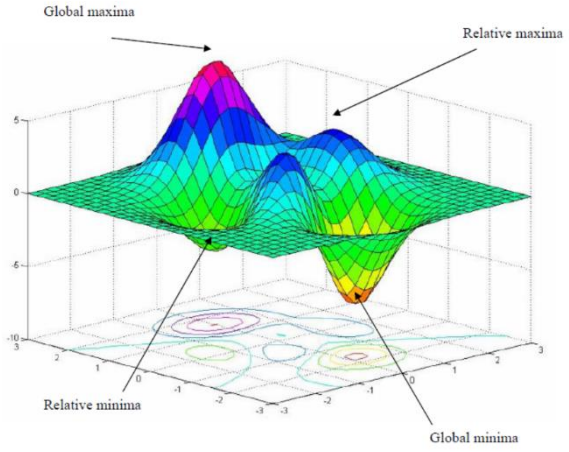
Consider $x = x^* = 2$

$$f'(x^*) = 240(x^*)^3 - 540(x^*)^2 + 240x^* = 240 \text{ at } x^* = 2$$

Since the second derivative is positive, the point $x = x^* = 2$ is a point of local minima with a minimum value of $f(x) = -11$

Functions of two variables

This concept may be easily extended to functions of multiple variables. Functions of two variables are best illustrated by contour maps, analogous to geographical maps. A contour is a line representing a constant value of $f(x)$ as shown in Fig.4. From this we can identify *maxima, minima* and *points of inflection.*



Necessary conditions

As can be seen in Fig. 4 and 5, perturbations from points of local minima in any direction result in an increase in the response function $f(x)$, i.e. the slope of the function is zero at this point of local minima. Similarly, at *maxima* and *points of inflection* as the slope is zero, the first derivatives of the function with respect to the variables are zero.

Which gives us $\frac{\partial f}{\partial x_1} = 0; \frac{\partial f}{\partial x_2} = 0$ at the stationary points, i.e., the gradient vector of $f(\mathbf{X})$, $\Delta_x f$

Sufficient conditions

Consider the following second order derivatives:

$$\frac{\partial^2 f}{\partial x_1^2}; \frac{\partial^2 f}{\partial x_2^2}; \frac{\partial^2 f}{\partial x_1 \partial x_2}$$

The Hessian matrix defined by **H** is made using the above second order derivatives.

$$\mathbf{H} = \begin{pmatrix} \dfrac{\partial^2 f}{\partial x_1^2} & \dfrac{\partial^2 f}{\partial x_1 \partial x_2} \\ \dfrac{\partial^2 f}{\partial x_1 \partial x_2} & \dfrac{\partial^2 f}{\partial x_2^2} \end{pmatrix}_{[x_1, x_2]}$$

a) If **H** is positive definite then the point $\mathbf{X} = [x_1, x_2]$ is a point of local minima.

b) If **H** is negative definite then the point $\mathbf{X} = [x_1, x_2]$ is a point of local maxima.

c) If **H** is neither then the point $\mathbf{X} = [x_1, x_2]$ is neither a point of maxima nor minima.

A square matrix is positive definite if all its eigen values are positive and it is negative definite if all its eigen values are negative. If some of the eigen values are positive and some negative then the matrix is neither positive definite or negative definite.

To calculate the eigen values λ of a square matrix then the following equation is solved.

$$|\mathbf{A} - \lambda \mathbf{I}| = 0$$

The above rules give the sufficient conditions for the optimization problem of two variables.

Locate the stationary points of $f(\mathbf{X})$ and classify them as relative maxima, relative minima or neither based on the rules discussed in the lecture.

$f(\mathbf{X}) = 2x_1^3/3 - 2x_1x_2 - 5x_1 + 2x_2^2 + 4x_2 + 5$

Solution

$$\Delta_x f = \begin{bmatrix} \frac{\partial f}{\partial x_1}(X^*) \\ \frac{\partial f}{\partial x_2}(X^*) \end{bmatrix} = \begin{bmatrix} 2x_1^2 - 2x_2 - 5 \\ -2x_1 + 4x_2 + 4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

From $\frac{\partial f}{\partial x_2}(X) = 0$, $x_1 = 2x_2 + 2$

From $\frac{\partial f}{\partial x_1}(X) = 0$

$$2(2x_2 + 2)^2 - 2x_2 - 5 = 0$$

$$8x_2^2 + 14x_2 + 3 = 0$$

$$(2x_2 + 3)(4x_2 + 1) = 0$$

$$x_2 = -3/2 \quad \text{or} \quad x_2 = -1/4$$

so the two stationary points are

$$X_1 = [-1, -3/2]$$

and

$$X_2 = [3/2, -1/4]$$

The Hessian of $f(\mathbf{X})$ is

$$\frac{\partial^2 f}{\partial x_1^2} = 4x_1; \frac{\partial^2 f}{\partial x_2^2} = 4; \frac{\partial^2 f}{\partial x_1 \partial x_2} = \frac{\partial^2 f}{\partial x_2 \partial x_1} = -2$$

$$\mathbf{H} = \begin{bmatrix} 4x_1 & -2 \\ -2 & 4 \end{bmatrix}$$

$$|\lambda \mathbf{I} - \mathbf{H}| = \begin{vmatrix} \lambda - 4x_1 & 2 \\ 2 & \lambda - 4 \end{vmatrix}$$

At X_1= [-1,-3/2],

$$|\lambda \mathbf{I} - \mathbf{H}| = \begin{vmatrix} \lambda + 4 & 2 \\ 2 & \lambda - 4 \end{vmatrix} = (\lambda + 4)(\lambda - 4) - 4 = 0$$

$$\lambda^2 - 16 - 4 = 0$$

$$\lambda^2 = 12$$

$$\lambda_1 = +\sqrt{12} \qquad \lambda_2 = -\sqrt{12}$$

Since one eigen value is positive and one negative, X_1 is neither a relative maximum nor a relative minimum.

At X_2 = [3/2,-1/4]

$$|\lambda \mathbf{I} - \mathbf{H}| = \begin{vmatrix} \lambda - 6 & 2 \\ 2 & \lambda - 4 \end{vmatrix} = (\lambda - 6)(\lambda - 4) - 4 = 0$$

$$\lambda^2 - 10\lambda + 20 = 0$$

$$\lambda_1 = 5 + \sqrt{5} \quad \lambda_2 = 5 - \sqrt{5}$$

Since both the eigen values are positive, X_2 is a local minimum.

Minimum value of f(x) is -0.375.

Optimization of Functions of Multiple Variables subject to Equality Constraints

Introduction

In the previous lecture we learnt the optimization of functions of multiple variables studied for unconstrained optimization. This is done with the aid of the gradient vector and the Hessian matrix. In this lecture we will learn the optimization of functions of multiple variables subjected to equality constraints using the method of constrained variation and the method of Lagrange multipliers.

Constrained optimization

A function of multiple variables, $f(x)$, is to be optimized subject to one or more equality constraints of many variables. These equality constraints, $g_j(x)$, may or may not be linear. The problem statement is as follows:

Maximize (or minimize) $f(\mathbf{X})$, subject to $g_j(\mathbf{X}) = 0$, j = 1, 2, … , m

where

$$X = \begin{Bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{Bmatrix}$$

with the condition that $m \leq n$; or else if $m > n$ then the problem becomes an over defined one and there will be no solution. Of the many available methods, the method of constrained variation and the method using Lagrange multipliers are discussed.

Solution by method of Constrained Variation

For the optimization problem defined above, let us consider a specific case with n = 2 and m=1 before we proceed to find the necessary and sufficient conditions for a general problem using Lagrange multipliers. The problem statement is as follows:

Minimize $f(x_1,x_2)$, subject to $g(x_1,x_2) = 0$

For $f(x_1,x_2)$ to have a minimum at a point $\mathbf{X}^* = [x_1^*,x_2^*]$, a necessary condition is that the total derivative of $f(x_1,x_2)$ must be zero at $[x_1^*,x_2^*]$.

$$df = \frac{\partial f}{\partial x_1} dx_1 + \frac{\partial f}{\partial x_2} dx_2 = 0 \qquad (1)$$

Since $g(x_1^*,x_2^*) = 0$ at the minimum point, variations dx_1 and dx_2 about the point $[x_1^*, x_2^*]$ must be *admissible variations*, i.e. the point lies on the constraint:

$$g(x_1^* + dx_1 , x_2^* + dx_2) = 0 \tag{2}$$

assuming dx_1 and dx_2 are small the Taylor series expansion of this gives us

$$g(x_1^*+dx_1 , x_2^*+dx_2) = g(x_1^*, x_2^*)+\frac{\partial g}{\partial x_1}(x_1^*,x_2^*)\, dx_1+\frac{\partial g}{\partial x_2}(x_1^*,x_2^*)\, dx_2 = 0 \tag{3}$$

or

$$dg = \frac{\partial g}{\partial x_1}dx_1+\frac{\partial g}{\partial x_2}dx_2 = 0 \text{ at } [x_1^*,x_2^*] \tag{4}$$

which is the condition that must be satisfied for all *admissible variations*.

Assuming $\partial g / \partial x_2 \neq 0$ (4) can be rewritten as

$$dx_2 = -\frac{\partial g / \partial x_1}{\partial g / \partial x_2}(x_1^*, x_2^*)dx_1 \tag{5}$$

which indicates that once variation along x_1 ($d\,x_1$) is chosen arbitrarily, the variation along x_2 ($d\,x_2$) is decided automatically to satisfy the condition for the *admissible variation*.

Substituting equation (5) *in* (1)we have:

$$df = \left(\frac{\partial f}{\partial x_1}-\frac{\partial g / \partial x_1}{\partial g / \partial x_2}\frac{\partial f}{\partial x_2}\right)\Bigg|_{(x_1^*, x_2^*)} dx_1 = 0 \tag{6}$$

The equation on the left hand side is called the constrained variation of f. Equation (5) has to be satisfied for all dx_1, hence we have

$$\left(\frac{\partial f}{\partial x_1}\frac{\partial g}{\partial x_2}-\frac{\partial f}{\partial x_2}\frac{\partial g}{\partial x_1}\right)\Bigg|_{(x_1^*, x_2^*)} = 0 \tag{7}$$

This gives us the necessary condition to have $[x_1^*, x2^*]$ as an extreme point (maximum or minimum)

Solution by method of Lagrange multipliers

Continuing with the same specific case of the optimization problem with $n = 2$ and $m = 1$ we define a quantity λ, called the *Lagrange multiplier* as

$$\lambda = -\left.\frac{\partial f / \partial x_2}{\partial g / \partial x_2}\right|_{(x_1^*, x_2^*)} \tag{8}$$

Using this in (6)

$$\left.\left(\frac{\partial f}{\partial x_1} + \lambda \frac{\partial g}{\partial x_1}\right)\right|_{(x_1^*, x_2^*)} = 0 \tag{9}$$

And (8) written as

$$\left.\left(\frac{\partial f}{\partial x_2} + \lambda \frac{\partial g}{\partial x_2}\right)\right|_{(x_1^*, x_2^*)} = 0 \tag{10}$$

Also, the constraint equation has to be satisfied at the extreme point

$$g(x_1, x_2)\big|_{(x_1^*, x_2^*)} = 0 \tag{11}$$

Hence equations (9) to (11) represent the necessary conditions for the point [x_1^*, $x2^*$] to be an extreme point.

Note that λ could be expressed in terms of $\partial g / \partial x_1$ as well and $\partial g / \partial x_1$ has to be non-zero. Thus, these necessary conditions require that at least one of the partial derivatives of $g(x_1, x_2)$ be non-zero at an extreme point.

The conditions given by equations (9) to (11) can also be generated by constructing a function ***L***, known as the Lagrangian function, as

$$L(x_1, x_2, \lambda) = f(x_1, x_2) + \lambda g(x_1, x_2) \tag{12}$$

Alternatively, treating $\boldsymbol{L}$ as a function of x_1, x_2 and λ, the necessary conditions for its extremum are given by

$$\frac{\partial L}{\partial x_1}(x_1, x_2, \lambda) = \frac{\partial f}{\partial x_1}(x_1, x_2) + \lambda \frac{\partial g}{\partial x_1}(x_1, x_2) = 0$$

$$\frac{\partial L}{\partial x_2}(x_1, x_2, \lambda) = \frac{\partial f}{\partial x_2}(x_1, x_2) + \lambda \frac{\partial g}{\partial x_2}(x_1, x_2) = 0 \qquad (13)$$
$$\frac{\partial L}{\partial \lambda}(x_1, x_2, \lambda) = g(x_1, x_2) = 0$$

The necessary and sufficient conditions for a general problem are discussed next.

Necessary conditions for a general problem

For a general problem with n variables and m equality constraints the problem is defined as shown earlier

Maximize (or minimize) $f(\mathbf{X})$, subject to $g_j(\mathbf{X}) = 0$, j = 1, 2, … , m

where $$\mathbf{X} = \begin{Bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{Bmatrix}$$

In this case the Lagrange function, $\boldsymbol{L}$, will have one Lagrange multiplier λ_j for each constraint $g_j(\mathrm{X})$ as

$$L(x_1, x_2, ..., x_n, \lambda_1, \lambda_2, ..., \lambda_m) = f(\mathbf{X}) + \lambda_1 g_1(\mathbf{X}) + \lambda_2 g_2(\mathbf{X}) + ... + \lambda_m g_m(\mathbf{X}) \qquad (14)$$

$\boldsymbol{L}$ is now a function of $n + m$ unknowns, $x_1, x_2, ..., x_n, \lambda_1, \lambda_2, ..., \lambda_m$, and the necessary conditions for the problem defined above are given by

$$\frac{\partial L}{\partial x_i} = \frac{\partial f}{\partial x_i}(\mathbf{X}) + \sum_{j=1}^{m} \lambda_j \frac{\partial g_j}{\partial x_i}(\mathbf{X}) = 0, \qquad i = 1, 2, ..., n \quad j = 1, 2, ..., m$$
$$\frac{\partial L}{\partial \lambda_j} = g_j(\mathbf{X}) = 0, \qquad j = 1, 2, ..., m \qquad (15)$$

which represent $n + m$ equations in terms of the $n + m$ unknowns, x_i and λ_j. The solution to this set of equations gives us

$$\mathbf{X} = \begin{Bmatrix} x_1^* \\ x_2^* \\ \vdots \\ x_n^* \end{Bmatrix} \quad \text{and} \quad \lambda^* = \begin{Bmatrix} \lambda_1^* \\ \lambda_2^* \\ \vdots \\ \lambda_m^* \end{Bmatrix} \qquad (16)$$

The vector $\mathbf{X}$ corresponds to the relative constrained minimum of $f(\mathbf{X})$ (subject to the verification of sufficient conditions).

Sufficient conditions for a general problem

A sufficient condition for $f(\mathbf{X})$ to have a relative minimum at $\mathbf{X}^*$ is that each root of the polynomial in ϵ, defined by the following determinant equation be positive.

$$\begin{vmatrix} L_{11}-\epsilon & L_{12} & \cdots & L_{1n} & g_{11} & g_{21} & \cdots & g_{m1} \\ L_{21} & L_{22}-\epsilon & & L_{2n} & g_{12} & g_{22} & & g_{m2} \\ \vdots & & \ddots & \vdots & \vdots & & \ddots & \vdots \\ L_{n1} & L_{n2} & \cdots & L_{nn}-\epsilon & g_{1n} & g_{2n} & \cdots & g_{mn} \\ g_{11} & g_{12} & \cdots & g_{1n} & 0 & \cdots & \cdots & 0 \\ g_{21} & g_{22} & & g_{2n} & \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots & \vdots & & & \vdots \\ g_{m1} & g_{m2} & \cdots & g_{mn} & 0 & \cdots & \cdots & 0 \end{vmatrix} = 0 \qquad (17)$$

where

$$L_{ij} = \frac{\partial^2 L}{\partial x_i \partial x_j}(\mathbf{X}^*, \lambda^*), \qquad \text{for } i = 1, 2, \ldots, n \quad j = 1, 2, \ldots, m$$
$$g_{pq} = \frac{\partial g_p}{\partial x_q}(\mathbf{X}^*), \qquad \text{where } p = 1, 2, \ldots, m \text{ and } q = 1, 2, \ldots, n \qquad (18)$$

Similarly, a sufficient condition for $f(\mathbf{X})$ to have a relative maximum at $\mathbf{X}^*$ is that each root of the polynomial in ϵ, defined by equation (17) be negative. If equation (17), on solving yields roots, some of which are positive and others negative, then the point $\mathbf{X}^*$ is neither a maximum nor a minimum.

Minimize

$f(\mathbf{X}) = -3x_1^2 - 6x_1x_2 - 5x_2^2 + 7x_1 + 5x_2$

Subject to $x_1 + x_2 = 5$

Solution

$g_1(\mathbf{X}) = x_1 + x_2 - 5 = 0$

$\mathbf{L}(x_1, x_2, ..., x_n, \lambda_1, \lambda_2, ..., \lambda_m) = f(\mathbf{X}) + \lambda_1 g_1(\mathbf{X}) + \lambda_2 g_2(\mathbf{X}) + ... + \lambda_m g_m(\mathbf{X})$ with $n = 2$ and $m = 1$

$$\mathbf{L} = -3x_1^2 - 6x_1x_2 - 5x_2^2 + 7x_1 + 5x_2 + \lambda_1(x_1 + x_2 - 5)$$

$$\frac{\partial \mathbf{L}}{\partial x_1} = -6x_1 - 6x_2 + 7 + \lambda_1 = 0$$

$$\Rightarrow x_1 + x_2 = \frac{1}{6}(7 + \lambda_1)$$

$$\Rightarrow 5 = \frac{1}{6}(7 + \lambda_1)$$

or $$\lambda_1 = 23$$

$$\frac{\partial \mathbf{L}}{\partial x_2} = -6x_1 - 10x_2 + 5 + \lambda_1 = 0$$

$$\Rightarrow 3x_1 + 5x_2 = \frac{1}{2}(5 + \lambda_1)$$

$$\Rightarrow 3(x_1 + x_2) + 2x_2 = \frac{1}{2}(5 + \lambda_1)$$

$$x_2 = \frac{-1}{2}$$

and, $$x_1 = \frac{11}{2}$$

Hence $\mathbf{X}^* = \left[\frac{-1}{2}, \frac{11}{2}\right]; \lambda^* = [23]$

$$\begin{pmatrix} L_{11} - \epsilon & L_{12} & g_{11} \\ L_{21} & L_{22} - \epsilon & g_{21} \\ g_{11} & g_{12} & 0 \end{pmatrix} = 0$$

$$L_{11} = \left.\frac{\partial^2 \mathbf{L}}{\partial x_1^2}\right|_{(\mathbf{X}^*, \lambda^*)} = -6$$

$$L_{12} = L_{21} = \left.\frac{\partial^2 \mathbf{L}}{\partial x_1 \partial x_2}\right|_{(\mathbf{X}^*, \lambda^*)} = -6$$

$$L_{22} = \left.\frac{\partial^2 \mathbf{L}}{\partial x_2^2}\right|_{(\mathbf{X}^*, \lambda^*)} = -10$$

$$g_{11} = \left.\frac{\partial g_1}{\partial x_1}\right|_{(\mathbf{X}^*, \lambda^*)} = 1$$

$$g_{12} = g_{21} = \left.\frac{\partial g_1}{\partial x_2}\right|_{(\mathbf{X}^*, \lambda^*)} = 1$$

The determinant becomes

$$\begin{pmatrix} -6 - \epsilon & -6 & 1 \\ -6 & -10 - \epsilon & 1 \\ 1 & 1 & 0 \end{pmatrix} = 0$$

or
$$(-6 - \epsilon)[-1] - (-6)[-1] + 1[-6 + 10 + \epsilon] = 0$$
$$\Rightarrow \epsilon = -2$$

Since ϵ is negative, $\mathbf{X}^*, \lambda^*$ correspond to a maximum.

Kuhn-Tucker Conditions

Introduction

In the previous lecture the optimization of functions of multiple variables subjected to equality constraints using the method of constrained variation and the method of Lagrange multipliers was dealt. In this lecture the Kuhn-Tucker conditions will be discussed with examples for a point to be a local optimum in case of a function subject to inequality constraints.

Kuhn-Tucker Conditions

It was previously established that for both an unconstrained optimization problem and an optimization problem with an equality constraint the first-order conditions are sufficient for a global optimum when the objective and constraint functions satisfy appropriate concavity/convexity conditions. The same is true for an optimization problem with inequality constraints.

The Kuhn-Tucker conditions are both necessary and sufficient if the objective function is concave and each constraint is linear or each constraint function is concave, i.e. the problems belong to a class called the *convex programming problems.*

Consider the following optimization problem:

Minimize $f(\mathbf{X})$ subject to $g_j(\mathbf{X}) \leq 0$ for $j = 1,2,\ldots,p$; where $\mathbf{X} = [x_1 \; x_2 \ldots x_n]$

Then the Kuhn-Tucker conditions for $\mathbf{X}^* = [x_1^* \; x_2^* \; \ldots x_n^*]$ to be a local minimum are

$$\begin{aligned} \frac{\partial f}{\partial x_i} + \sum_{j=1}^{m} \lambda_j \frac{\partial g}{\partial x_i} = 0 \qquad & i = 1,2,\ldots,n \\ \lambda_j g_j = 0 \qquad & j = 1,2,\ldots,m \\ g_j \leq 0 \qquad & j = 1,2,\ldots,m \\ \lambda_j \geq 0 \qquad & j = 1,2,\ldots,m \end{aligned} \tag{1}$$

In case of minimization problems, if the constraints are of the form $g_j(\mathbf{X}) \geq 0$, then λ_j have to be nonpositive in (1). On the other hand, if the problem is one of maximization with the constraints in the form $g_j(\mathbf{X}) \geq 0$, then λ_j have to be nonnegative.

It may be noted that sign convention has to be strictly followed for the Kuhn-Tucker conditions to be applicable.

Example 1

Minimize $f = x_1^2 + 2x_2^2 + 3x_3^2$ subject to the constraints

$$g_1 = x_1 - x_2 - 2x_3 \leq 12$$
$$g_2 = x_1 + 2x_2 - 3x_3 \leq 8$$

using Kuhn-Tucker conditions.

Solution:

The Kuhn-Tucker conditions are given by

a) $\dfrac{\partial f}{\partial x_i} + \lambda_1 \dfrac{\partial g_1}{\partial x_i} + \lambda_2 \dfrac{\partial g_2}{\partial x_i} = 0$

i.e.

$$2x_1 + \lambda_1 + \lambda_2 = 0 \quad (2)$$
$$4x_2 - \lambda_1 + 2\lambda_2 = 0 \quad (3)$$
$$6x_3 - 2\lambda_1 - 3\lambda_2 = 0 \quad (4)$$

b) $\lambda_j g_j = 0$

i.e.

$$\lambda_1(x_1 - x_2 - 2x_3 - 12) = 0 \quad (5)$$
$$\lambda_2(x_1 + 2x_2 - 3x_3 - 8) = 0 \quad (6)$$

c) $g_j \leq 0$

i.e.,

$$x_1 - x_2 - 2x_3 - 12 \le 0 \quad (7)$$
$$x_1 + 2x_2 - 3x_3 - 8 \le 0 \quad (8)$$

d) $\lambda_j \ge 0$

i.e.,

$$\lambda_1 \ge 0 \quad (9)$$
$$\lambda_2 \ge 0 \quad (10)$$

From (5) either $\lambda_1 = 0$ or, $x_1 - x_2 - 2x_3 - 12 = 0$

Case 1: $\lambda_1 = 0$

From *(2), (3) and (4)* we have $x_1 = x_2 = -\lambda_2 / 2$ and $x_3 = \lambda_2 / 2$.

Using these in *(6)* we get $\lambda_2^2 + 8\lambda_2 = 0, \therefore \lambda_2 = 0$ *or* -8

From *(10)*, $\lambda_2 \ge 0$, therefore, $\lambda_2 = 0$, $\mathbf{X}^* = [\,0, 0, 0\,]$, this solution set satisfies all of *(6)* to *(9)*

Case 2: $x_1 - x_2 - 2x_3 - 12 = 0$

Using *(2), (3) and (4),* we have $\frac{-\lambda_1 - \lambda_2}{2} - \frac{\lambda_1 - 2\lambda_2}{4} - \frac{2\lambda_1 + 3\lambda_2}{3} - 12 = 0$ or,

$17\lambda_1 + 12\lambda_2 = -144$. But conditions *(9)* and *(10)* give us $\lambda_1 \ge 0$ and $\lambda_2 \ge 0$ simultaneously, which cannot be possible with $17\lambda_1 + 12\lambda_2 = -144$.

Hence the solution set for this optimization problem is $\mathbf{X}^* = [\,0\ 0\ 0\,]$

Example 2

Minimize $f = x_1^2 + x_2^2 + 60x_1$ subject to the constraints

$g_1 = x_1 - 80 \geq 0$
$g_2 = x_1 + x_2 - 120 \geq 0$

using Kuhn-Tucker conditions.

Solution

The Kuhn-Tucker conditions are given by

a) $\frac{\partial f}{\partial x_i} + \lambda_1 \frac{\partial g_1}{\partial x_i} + \lambda_2 \frac{\partial g_2}{\partial x_i} + \lambda_3 \frac{\partial g_3}{\partial x_i} = 0$

i.e.

$$2x_1 + 60 + \lambda_1 + \lambda_2 = 0 \quad (11)$$
$$2x_2 + \lambda_2 = 0 \quad (12)$$

b) $\lambda_j g_j = 0$

i.e.

$$\lambda_1(x_1 - 80) = 0 \quad (13)$$
$$\lambda_2(x_1 + x_2 - 120) = 0 \quad (14)$$

c) $g_j \leq 0$

i.e.,

$$x_1 - 80 \geq 0 \quad (15)$$
$$x_1 + x_2 + 120 \geq 0 \quad (16)$$

d) $\lambda_j \leq 0$

i.e.,

$$\lambda_1 \le 0 \tag{17}$$
$$\lambda_2 \le 0 \tag{18}$$

From (13) either $\lambda_1 = 0$ or, $(x_1 - 80) = 0$

Case 1: $\lambda_1 = 0$

From *(11) and (12)* we have $x_1 = -\lambda_2/2 - 30$ and $x_2 = -\lambda_2/2$

Using these in *(14)* we get $\lambda_2(\lambda_2 - 150) = 0$; $\therefore \lambda_2 = 0$ *or* -150

Considering $\lambda_2 = 0$, $\mathbf{X}^* = [\,30, 0]$.

But this solution set violates *(15)* and *(16)*

For $\lambda_2 = -150$, $\mathbf{X}^* = [\,45, 75]$.

But this solution set violates *(15)* .

Case 2: $(x_1 - 80) = 0$

Using $x_1 = 80$ in *(11)* and *(12)*, we have

$$\lambda_2 = -2x_2$$
$$\lambda_1 = 2x_2 - 220 \tag{19}$$

Substitute (19) in (14), we have

$$-2x_2(x_2 - 40) = 0 .$$

For this to be true, either $x_2 = 0$ *or* $x_2 - 40 = 0$

For $x_2 = 0$, $\lambda_1 = -220$. This solution set violates *(15)* and *(16)*

For $x_2 - 40 = 0$, $\lambda_1 = -140$ *and* $\lambda_2 = -80$. This solution set is satisfying all equations from (15) to (19) and hence the desired. Therefore, the solution set for this optimization problem is $\mathbf{X}^* = [\,80\ 40\,]$.